

MongoDB Library Exercise

1. Create Operation

```
ecommerceDB> use library
switched to db library
library> db.createCollection("books")
{ ok: 1 }
library> db.books.insertMany([
|   { title: "The Great Gatsby", author: "F. Scott Fitzgerald", published_year: 1925 },
|   { title: "Harry Potter and the Sorcerer's Stone", author: "J.K. Rowling", published_year: 1997 },
|   { title: "The Catcher in the Rye", author: "J.D. Salinger", published_year: 1951 },
|   { title: "To Kill a Mockingbird", author: "Harper Lee", published_year: 1960 },
|   { title: "1984", author: "George Orwell", published_year: 1949 },
|   { title: "MongoDB Basics", author: "John Doe", published_year: 2022 },
|   { title: "NoSQL for Beginners", author: "Jane Smith", published_year: 2021 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f64aca086823a248abc117'),
    '1': ObjectId('69f64aca086823a248abc118'),
    '2': ObjectId('69f64aca086823a248abc119'),
    '3': ObjectId('69f64aca086823a248abc11a'),
    '4': ObjectId('69f64aca086823a248abc11b'),
    '5': ObjectId('69f64aca086823a248abc11c'),
    '6': ObjectId('69f64aca086823a248abc11d')
  }
}
library> |
```

2. Read Operation

```
library> db.books.find().pretty()
[
  {
    _id: ObjectId('69f64aca086823a248abc117'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    published_year: 1925
  },
  {
    _id: ObjectId('69f64aca086823a248abc118'),
    title: 'Harry Potter and the Sorcerer's Stone',
    author: 'J.K. Rowling',
    published_year: 1997
  },
  {
    _id: ObjectId('69f64aca086823a248abc119'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 1951
  },
  {
    _id: ObjectId('69f64aca086823a248abc11a'),
    title: 'To Kill a Mockingbird',
    author: 'Harper Lee',
    published_year: 1960
  },
  {
    _id: ObjectId('69f64aca086823a248abc11b'),
    title: '1984',
    author: 'George Orwell',
    published_year: 1949
  },
  {
    _id: ObjectId('69f64aca086823a248abc11c'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022
  },
  {
    _id: ObjectId('69f64aca086823a248abc11d'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021
  }
]
library> |
```

```
library> db.books.find({ author: "J.K. Rowling" })
[
  {
    _id: ObjectId('69f64aca086823a248abc118'),
    title: "Harry Potter and the Sorcerer's Stone",
    author: 'J.K. Rowling',
    published_year: 1997
  }
]
library>
```

3. Update Operation

```
]
library> db.books.find().sort({ published_year: 1 }).limit(1)
[
  {
    _id: ObjectId('69f64aca086823a248abc117'),
    title: 'The Great Gatsby',
    author: 'F. Scott Fitzgerald',
    published_year: 1925
  }
]
library>
```

```
]
library> db.books.updateOne(
|   { title: "The Catcher in the Rye" },
|   { $set: { published_year: 2024 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
library>
```

4. Delete Operation

```
library> db.books.updateMany(  
|   {},  
|   { $set: { genre: "Mystery" } }  
| )  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 7,  
  modifiedCount: 7,  
  upsertedCount: 0  
}  
library>
```

5. Advanced Query

```
library> db.books.deleteOne({ title: "1984" })
{ acknowledged: true, deletedCount: 1 }
library> db.books.deleteMany({ published_year: { $lt: 2000 } })
{ acknowledged: true, deletedCount: 3 }
library> |
```

```
library> db.books.find()
| .sort({ published_year: -1 })
| .limit(3)
[
  {
    _id: ObjectId('69f64aca086823a248abc119'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2024,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f64aca086823a248abc11c'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f64aca086823a248abc11d'),
    title: 'NoSQL for Beginners',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
library>
```

```
library> db.books.find({
|   $or: [
|     { title: /MongoDB/i },
|     { title: /NoSQL/i }
|   ]
| })
[
|   {
|     _id: ObjectId('69f64aca086823a248abc11c'),
|     title: 'MongoDB Basics',
|     author: 'John Doe',
|     published_year: 2022,
|     genre: 'Mystery'
|   },
|   {
|     _id: ObjectId('69f64aca086823a248abc11d'),
|     title: 'NoSQL for Beginners',
|     author: 'Jane Smith',
|     published_year: 2021,
|     genre: 'Mystery'
|   }
| ]
library>
```

Conclusion:

This exercise demonstrates MongoDB CRUD operations including create, read, update, delete, and advanced queries using mongosh.